Geometric Intuition for Lagrange Multipliers

Key Concepts

1. Gradients and Level Curves

For a function $f(x, y)$, the gradient ∇f at a point is **perpendicular** to the level curve $f(x, y) = k$ passing through that point. This is because the directional derivative in any tangent direction $\mathbf{v}$ satisfies:

$$\nabla f \cdot \mathbf{v} = 0.$$

2. Tangency and Parallel Gradients

When the level curve $f(x, y) = k$ is **tangent** to the constraint $g(x, y) = c$ at a point (x^*, y^*):

- The gradients ∇f and ∇g must both be perpendicular to the shared tangent line.
- This implies $\nabla f \parallel \nabla g$, so:

$$\nabla f = \lambda \nabla g \quad \text{for some scalar } \lambda.$$

3. Example

Maximize $f(x, y) = x + y$ subject to $g(x, y) = x^2 + y^2 = 1$:

- The maximum occurs where the line $x + y = k$ is tangent to the circle $x^2 + y^2 = 1$.
- At the tangent point $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$:

$$\nabla f = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \nabla g = \begin{bmatrix} 2x \\ 2y \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ \sqrt{2} \end{bmatrix}.$$

- Clearly, $\nabla f = \lambda \nabla g$ with $\lambda = \frac{1}{\sqrt{2}}$.

4. Why This Matters

If ∇f and ∇g are not parallel at a point on $g(x, y) = c$, you could move along the constraint to increase/decrease f. Only when gradients are parallel (level curves tangent) is further improvement "blocked," making it a potential extremum.

Summary

Tangency of level curves $\implies \nabla f \parallel \nabla g \implies$ Lagrange multiplier condition

The Gradient Alignment Condition

From our earlier discussion, we know that at the extremum of $f(x)$ subject to the constraint $g(x) = c$:

$$\nabla f = \lambda \nabla g,$$

where λ is the Lagrange multiplier. This means that the gradients of f and g are parallel at the extremum.

However, this condition alone does not enforce the constraint $g(x) = c$. To incorporate the constraint into the optimization problem, we proceed as follows.

Introducing the Constraint

To ensure that the solution satisfies $g(x) = c$, we introduce the term $g(x) - c$, which measures how much the constraint is violated. At the solution, this term must equal zero:

$$g(x) - c = 0.$$

Constructing the Lagrangian

We combine the objective function $f(x)$ and the constraint $g(x) - c$ into a single function called the **Lagrangian**:

$$\mathcal{L}(x, \lambda) = f(x) - \lambda(g(x) - c).$$

Here:

- $f(x)$: The original objective function to be optimized.

- $-\lambda(g(x) - c)$: A penalty term that incorporates the constraint. The multiplier λ adjusts the weight of the constraint in the optimization.

Why Does This Work?

The Lagrangian encodes both the gradient alignment condition and the constraint satisfaction condition. Let's see why:

(a) Gradient Condition

Take the partial derivatives of $\mathcal{L}$ with respect to the variables x:

$$\frac{\partial \mathcal{L}}{\partial x} = \nabla f - \lambda \nabla g.$$

Setting $\frac{\partial \mathcal{L}}{\partial x} = 0$ gives:

$$\nabla f = \lambda \nabla g.$$

This is exactly the gradient alignment condition we derived earlier!

(b) Constraint Satisfaction

Take the partial derivative of $\mathcal{L}$ with respect to λ:

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(g(x) - c).$$

Setting $\frac{\partial \mathcal{L}}{\partial \lambda} = 0$ gives:

$$g(x) - c = 0 \quad \text{or equivalently} \quad g(x) = c.$$

This ensures that the constraint is satisfied.

Summary

By constructing the Lagrangian $\mathcal{L}(x, \lambda) = f(x) - \lambda(g(x) - c)$, we achieve two things:

1. The gradient alignment condition ($\nabla f = \lambda \nabla g$) is enforced by setting $\frac{\partial \mathcal{L}}{\partial x} = 0$.
2. The constraint $g(x) = c$ is enforced by setting $\frac{\partial \mathcal{L}}{\partial \lambda} = 0$.

Thus, solving $\nabla \mathcal{L} = 0$ simultaneously enforces both conditions, ensuring that we find the extrema of $f(x)$ subject to the constraint $g(x) = c$.

Final Boxed Answer

The Lagrangian $\mathcal{L}(x, \lambda) = f(x) - \lambda(g(x) - c)$ is constructed so that:

1. Setting $\frac{\partial \mathcal{L}}{\partial x} = 0$ enforces $\nabla f = \lambda \nabla g$ (gradient alignment).
2. Setting $\frac{\partial \mathcal{L}}{\partial \lambda} = 0$ enforces $g(x) = c$ (constraint satisfaction).

This ensures that the solution satisfies both conditions for constrained optimization:

$$\boxed{\mathcal{L}(x, \lambda) = f(x) - \lambda(g(x) - c)}$$

Example 1: Maximizing Volume of a Cylinder with Surface Area Constraint

Problem: Find the dimensions of a cylinder (radius r, height h) with maximum volume, given a fixed surface area $S = 12\pi$.

Solution:

- **Objective Function**: Volume $V = \pi r^2 h$.
- **Constraint**: Surface area $2\pi r^2 + 2\pi r h = 12\pi$.
- **Lagrangian**:

$$\mathcal{L}(r, h, \lambda) = \pi r^2 h - \lambda(2\pi r^2 + 2\pi r h - 12\pi).$$

- **Partial Derivatives**:

$$\frac{\partial \mathcal{L}}{\partial r} = 2\pi r h - \lambda(4\pi r + 2\pi h) = 0 \quad (1),$$

$$\frac{\partial \mathcal{L}}{\partial h} = \pi r^2 - \lambda(2\pi r) = 0 \quad (2),$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(2\pi r^2 + 2\pi r h - 12\pi) = 0 \quad (3).$$

- **Solve**:
  - From (2): $\pi r^2 = 2\pi r \lambda \implies r = 2\lambda$.
  - Substitute $\lambda = r/2$ into (1):

$$2\pi r h = \frac{r}{2}(4\pi r + 2\pi h) \implies 4h = 4r + 2h \implies h = 2r.$$

– Substitute $h = 2r$ into (3):

$$2\pi r^2 + 2\pi r(2r) = 12\pi \implies 6r^2 = 12 \implies r = \sqrt{2},\ h = 2\sqrt{2}.$$

- **Conclusion**: Maximum volume is achieved when $r = \sqrt{2}$, $h = 2\sqrt{2}$, and $V = 4\sqrt{2}\pi$.

Example 2: Minimizing Distance from a Point to a Plane

Problem: Find the point on the plane $2x + 3y + z = 6$ closest to the origin.

Solution:

- **Objective Function**: Distance squared $f(x, y, z) = x^2 + y^2 + z^2$.
- **Constraint**: Plane equation $2x + 3y + z = 6$.
- **Lagrangian**:

$$\mathcal{L}(x, y, z, \lambda) = x^2 + y^2 + z^2 - \lambda(2x + 3y + z - 6).$$

- **Partial Derivatives**:

$$\frac{\partial \mathcal{L}}{\partial x} = 2x - 2\lambda = 0 \implies x = \lambda \quad (1),$$
$$\frac{\partial \mathcal{L}}{\partial y} = 2y - 3\lambda = 0 \implies y = \frac{3}{2}\lambda \quad (2),$$
$$\frac{\partial \mathcal{L}}{\partial z} = 2z - \lambda = 0 \implies z = \frac{\lambda}{2} \quad (3),$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(2x + 3y + z - 6) = 0 \quad (4).$$

- **Solve**:

– Substitute (1), (2), (3) into (4):

$$2\lambda + 3\left(\frac{3}{2}\lambda\right) + \frac{\lambda}{2} = 6 \implies 7\lambda = 6 \implies \lambda = \frac{6}{7}.$$

– Critical point: $x = \frac{6}{7}$, $y = \frac{9}{7}$, $z = \frac{3}{7}$.

- **Conclusion**: Closest point is $\left(\frac{6}{7}, \frac{9}{7}, \frac{3}{7}\right)$, distance $\sqrt{\frac{18}{7}}$.

Example 3: Cobb-Douglas Production Function with Budget Constraint

Problem: A firm produces output $Q = K^{1/3}L^{2/3}$, where K is capital and L is labor. Maximize Q subject to a budget constraint $2L + K = 12$.

Solution:

- **Objective Function**: $Q = K^{1/3}L^{2/3}$.
- **Constraint**: $2L + K = 12$.
- **Lagrangian**:

$$\mathcal{L}(K, L, \lambda) = K^{1/3}L^{2/3} - \lambda(2L + K - 12).$$

- **Partial Derivatives**:

$$\frac{\partial \mathcal{L}}{\partial K} = \frac{1}{3}K^{-2/3}L^{2/3} - \lambda = 0 \quad (1),$$
$$\frac{\partial \mathcal{L}}{\partial L} = \frac{2}{3}K^{1/3}L^{-1/3} - 2\lambda = 0 \quad (2),$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(2L + K - 12) = 0 \quad (3).$$

- **Solve**:
  - From (1) and (2):

$$\frac{1}{3}\left(\frac{L}{K}\right)^{2/3} = \lambda, \quad \frac{2}{3}\left(\frac{K}{L}\right)^{1/3} = 2\lambda.$$

    Divide the two equations:

$$\frac{1}{2}\left(\frac{L}{K}\right) = \frac{1}{2} \implies L = K.$$

  - Substitute $L = K$ into (3):

$$2K + K = 12 \implies K = 4,\ L = 4.$$

- **Conclusion**: Maximum output $Q = 4^{1/3} \cdot 4^{2/3} = 4$.